

CT & CBCT



- ▶ Computed Tomography (CT): is a well-accepted imaging modality for evaluation of the **entire body**.
- ▶ Computed Tomography (CT) Scan Machines
Uses X- rays, a powerful form of
Electromagnetic Radiation.

- ▶ CT has undergone several evolutions and nowadays **multi- detectors CT** scanners have been evolved which have better application in clinical field.



Applications of CT

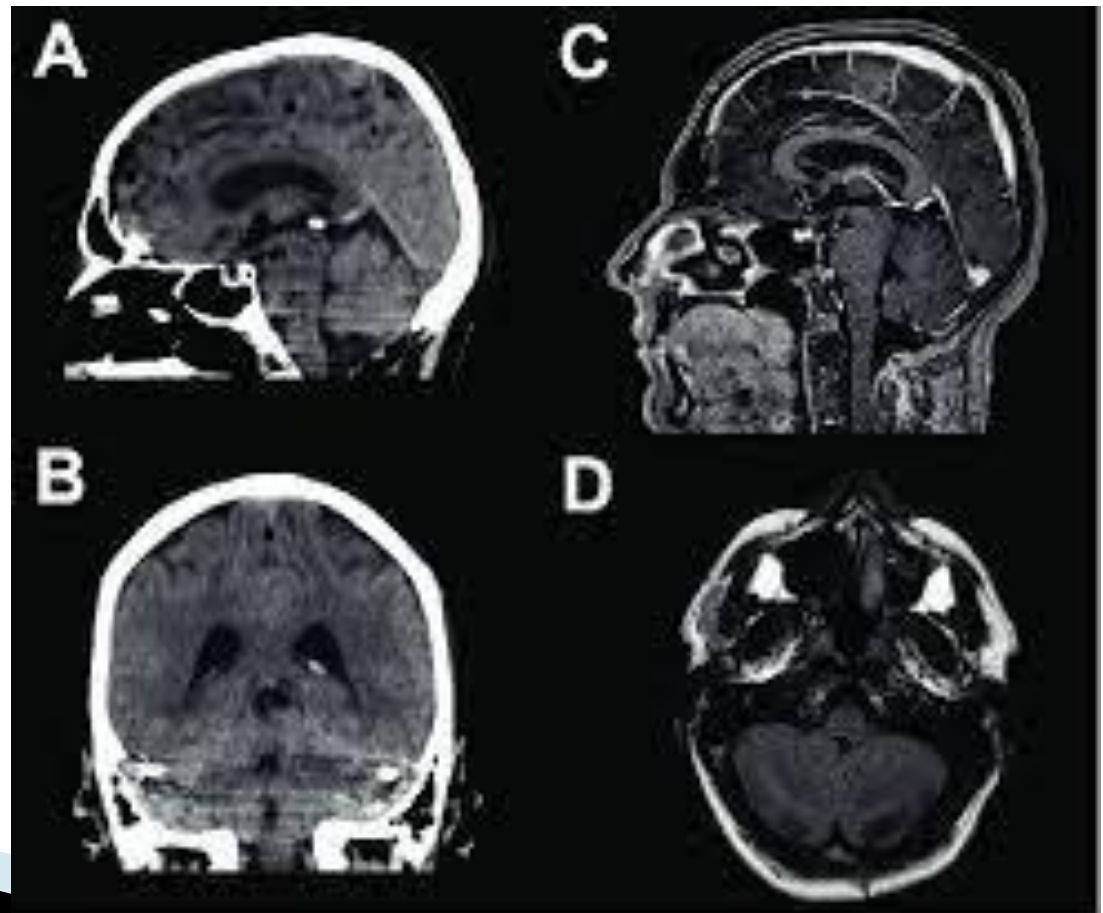
MDCT imaging has several applications in the diagnosis and treatment of dentomaxillofacial diseases

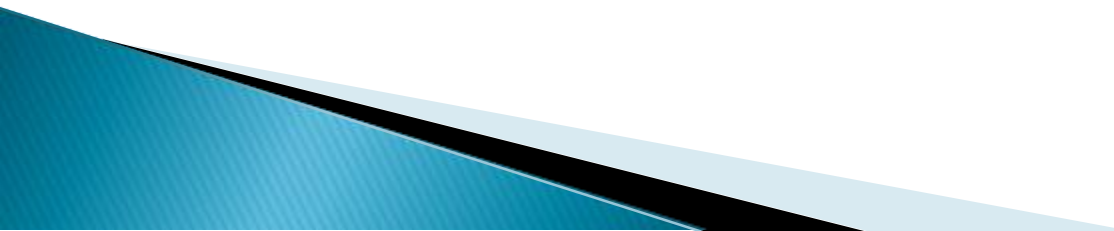
- ▶ Midfacial and mandibular **trauma** •
- ▶ **Developmental anomalies** of the craniofacial skeleton
- ▶ Benign intraosseous **cysts and neoplasms of the jaws**

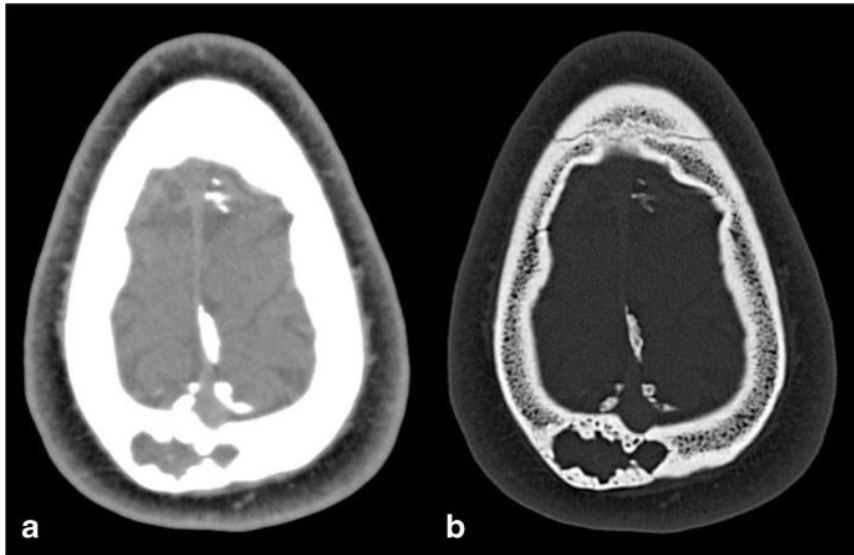
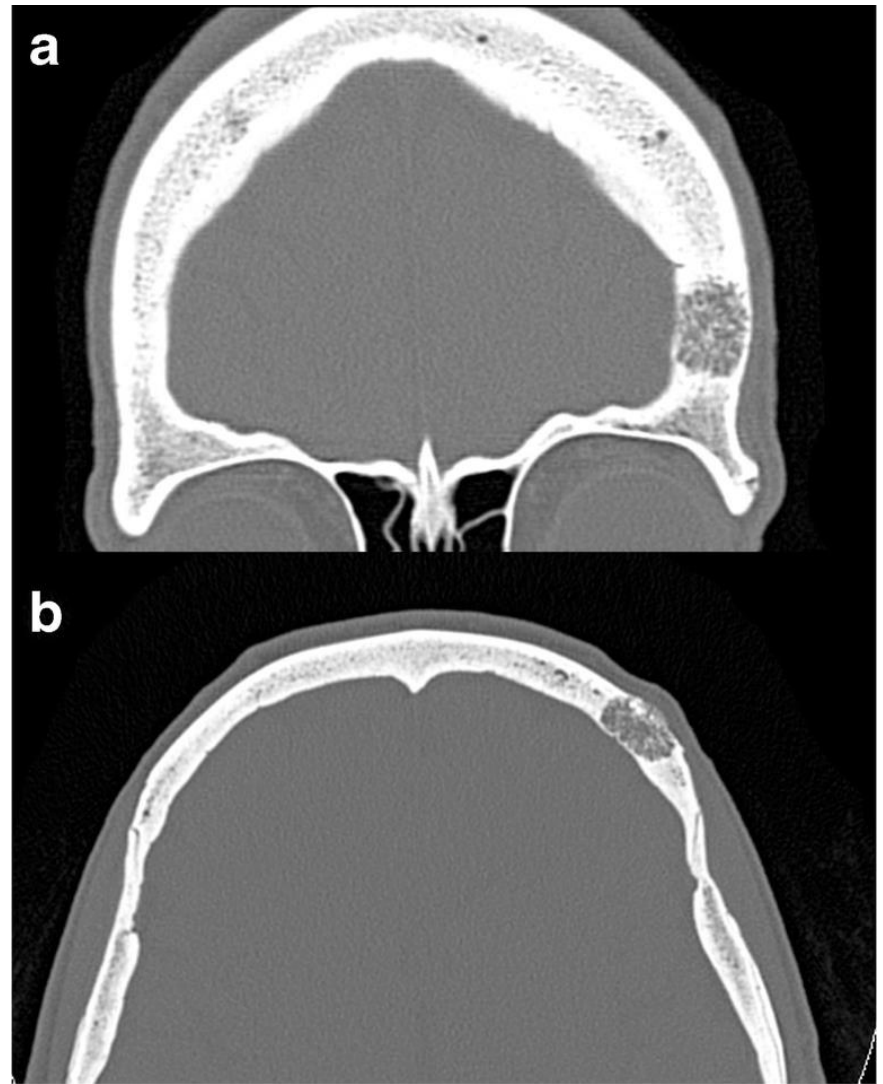
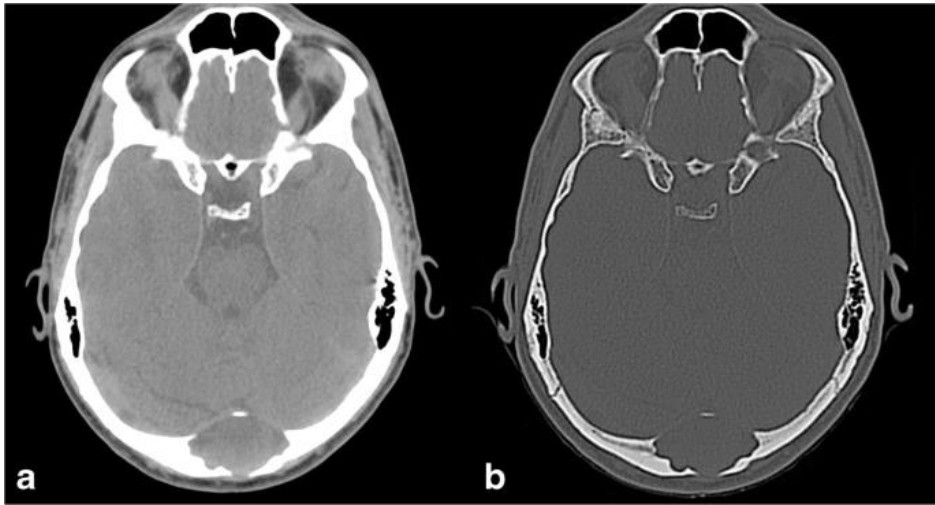
- ▶ Benign and malignant neoplasms that originate in, or extend into, the orofacial soft tissues •
- ▶ Soft-tissue cysts

Advantages of CT

- ▶ It provides **axial, coronal and sagittal** views of the tissue.

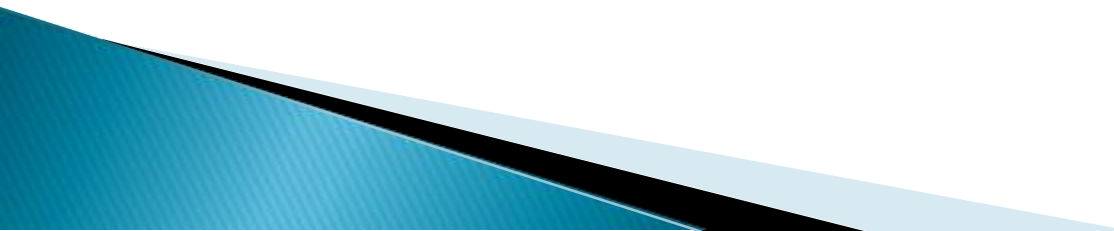


- ▶ • It shows anatomically **precise location** of the lesion and extent.
 - ▶ • The structures of the **soft tissues** both normal and pathological are clearly displayed
 - ▶ • Because the image that is produced is formulated by the computer, **areas of interest may be selectively viewed and enlarged** by using computer programmers.
 - ▶ • As the information is **stored in the computer** it can be viewed any time in the absence of patient.
 - ▶ • Image can be **manipulated**.
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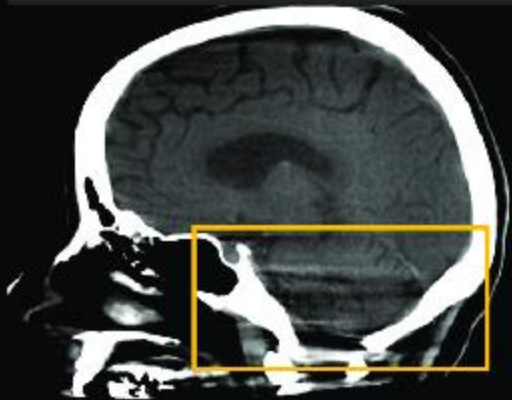


Disadvantages of CT

Is sophisticated, costly and expensive to maintain.

- ▶ • Very high-density materials such as metal bullets and dental restorations produce **severe artefacts on CT scan**, which makes the interpretation difficult.
 - ▶ • There is an **inherent risk** associated with the contrast medium.
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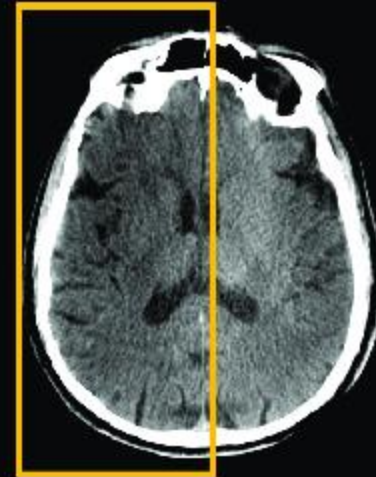
Beam Hardening



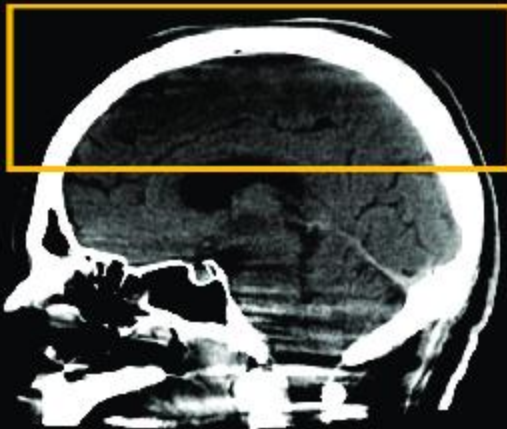
Undersampling



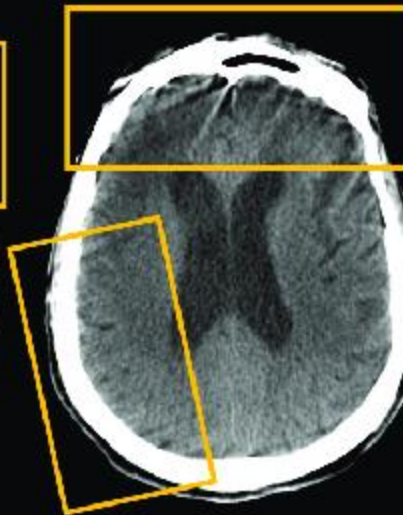
Inhomogeneity



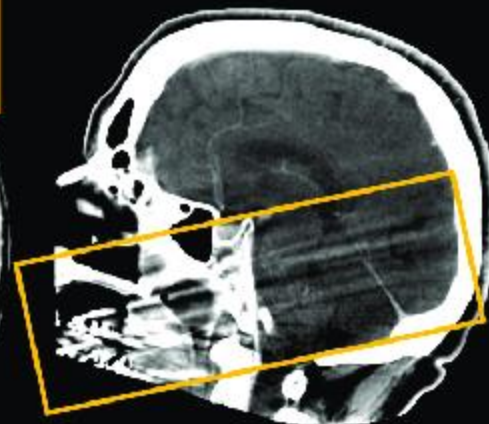
Cone-beam effect



Patient Motion

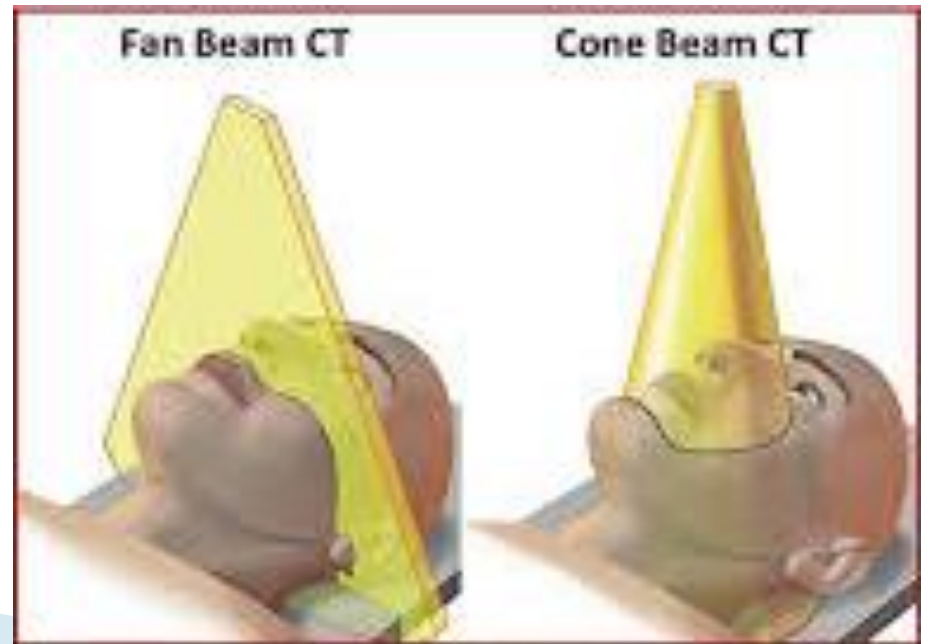
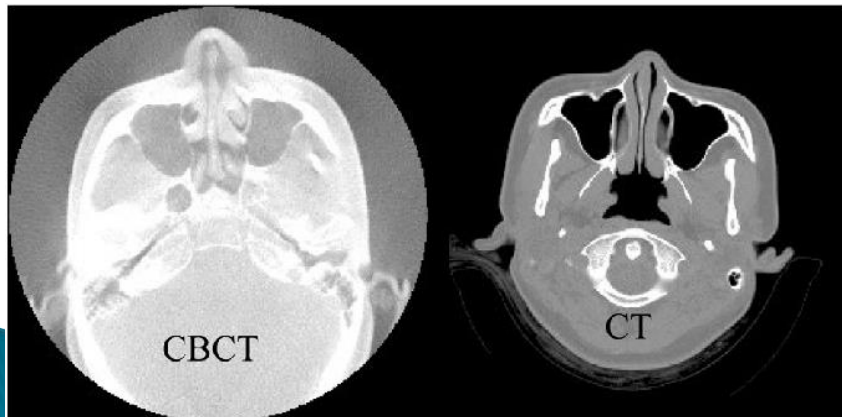


Metal Artifacts



Cone-beam computed tomography (CBCT)

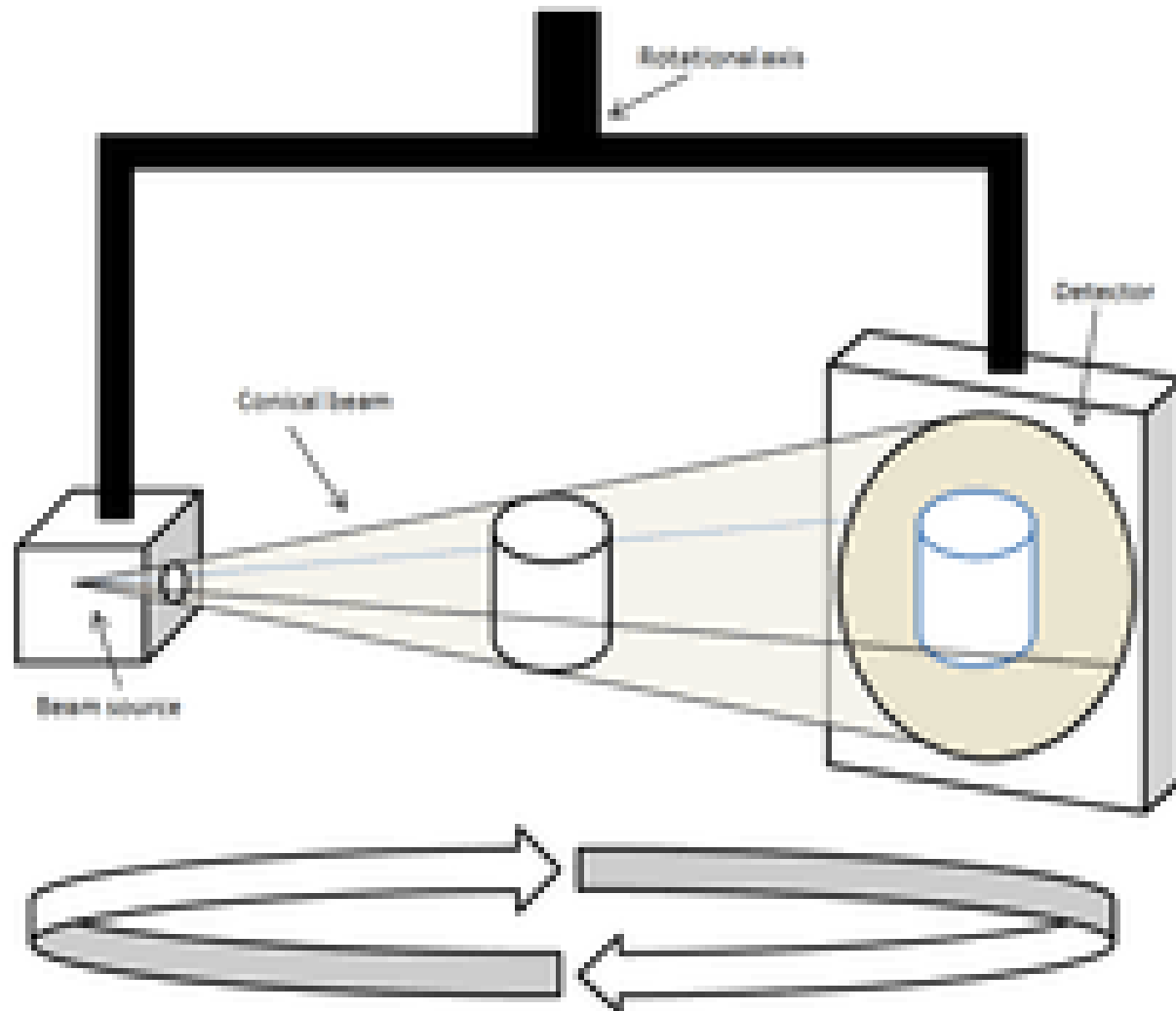
- ▶ Is a new medical imaging technique that **generates 3D images**.
- ▶ This imaging technique is based on a cone shaped X-ray beam centered on a 2D detector that performs **one rotation around the object**, producing a series of 2D images.



Components of CBCT

- ▶ CBCT is composed of X-ray source with a rotating gantry. A divergent pyramidal or cone-shaped source of ionizing radiation is directed through the middle of the region of interest .



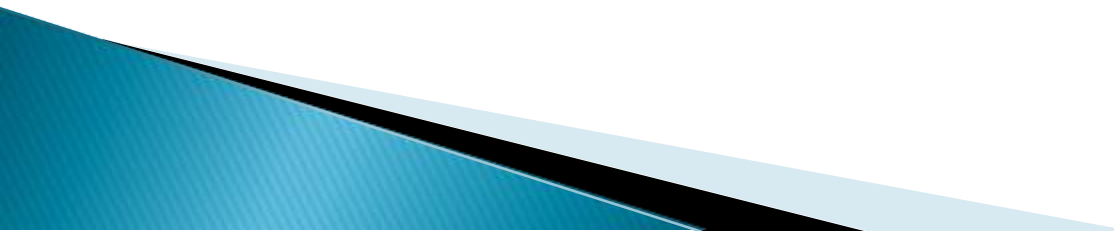




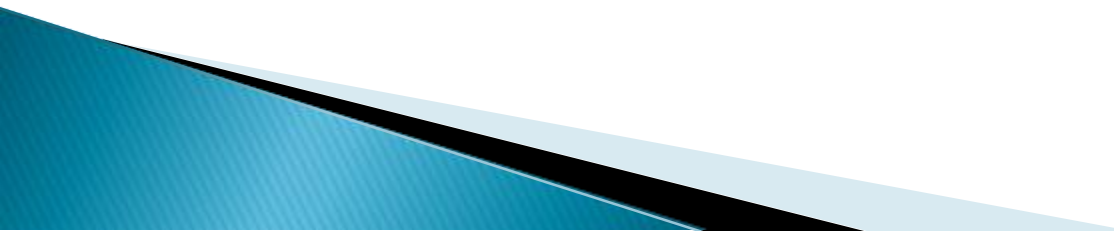
- ▶ CBCT scanner utilizes a 2D X-ray detector on the opposite side, which allows for a **single rotation of the gantry to generate a scan.**
- ▶ Computer to display the images



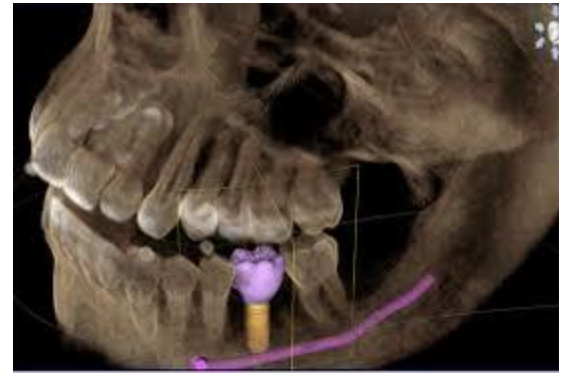
Advantages

- ▶ • Rapid scan time
 - ▶ • Beam limitation
 - ▶ • Image accuracy
 - ▶ • Reduced patient radiation dose
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Disadvantages

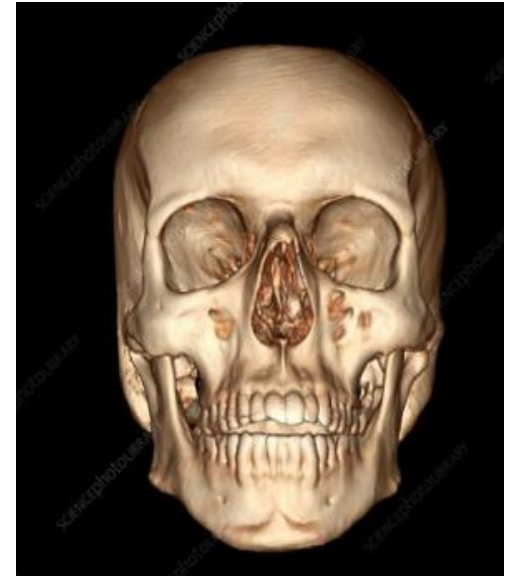
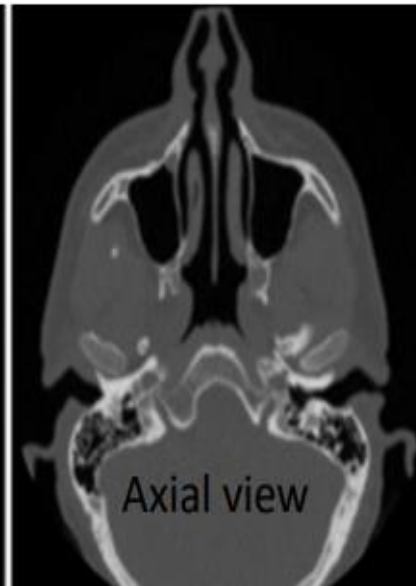
- ▶ • Extinction artefacts
 - ▶ • Beam hardening artefacts
 - ▶ • Ring artefacts
 - ▶ • Motion artefacts (misalignment artefacts)
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Application of CBCT



- ▶ Implant imaging
- ▶ orthodontic and orthognathic surgeries
- ▶ endodontics
- ▶ Periodontics
- ▶ TMG diagnosis
- ▶ Maxillary sinus assessment
- ▶ Impacted teeth (Third molar and canine)

Views interpretations and Anatomy



FOV

